

Dynamical Friction and Momentum Evolution

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1 Introduction

Dynamical Friction is a phenomenon wherein the momentum of an object is lost through gravitational interaction with other bodies. An example can be seen when two clouds of particles are passing through one another. Through the mutual attraction of the bodies with each other, the two clouds will pass by each other and exchange momentum. If say, one cloud has more massive particles, then the more massive cloud will slow down and the lighter cloud will speed up as a result of this interaction.

This example can be studied with the interaction of the galaxies: the Milky Way, M31, and M33 as the galaxies are just a conglomeration of stars and dark matter "particles" (at least for the purpose of simulation).

2 Topics of Study

Through the use of the simulation produced by van der Marel et al. (2012), I can track the location of particles at particular "snapshots" or time frames to see the evolution of this system.

1. I would like to analyze the effects of dynamical friction by determining the center of mass velocity as a function of time for the Milky Way, M31, and M33.

2. I would like to determine the momentum exchange of each pass of the galaxy merger, by determining the total translational momentum of each system and how it evolves in time.

3. I would like to track the evolution of the angular momentum in time as a result of this collision, by determining each system's total angular momentum.

3 References

[1] van der Marel, R. P., Besla, G., Cox, T. J., Sohn, S. T., Anderson, J. 2012, ApJ, 753, 9